

Ben Bioren

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EDUCATION

University of Washington

BS in Computer Science, BA in Mathematics. Robinson Scholar. 3.99 GPA

Seattle, Wa

Expected Grad. June 2027

- Relevant Coursework: Probability & Statistics, Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Distributed Systems, Linear Algebra, Discrete Modeling
- Lead Teaching Assistant: Calculus Series (x7), Teaching Assistant: Data Structures & Algorithms (x2)

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, HTML, CSS, Swift, R, SQL, Bash, Rust, Go, MATLAB

Technologies: React, React Native, Node.js, Flask, Django, FastAPI, SwiftUI, JUnit, Next.js, Express.js, Gradio

Developer Tools: Git, Docker, Kubernetes, Azure, AWS (Lambda, S3, EC2), Firebase, Pinecone, Hugging Face

Libraries: NumPy, Pandas, PyTorch, TensorFlow, Keras, SciPy, Matplotlib, SciKit, OpenCV, Selenium, BeautifulSoup

EXPERIENCE

Software Engineering Intern

June 2025 – September 2025

Levlepath (Series B Startup)

San Francisco, CA

- Optimized AI pipeline for robust search-intent extraction, reducing average completion time by 5s (33%) across queries
- Fine-tuned SDXL-based text-to-image diffusion models to enhance manufacturability metrics by 30% during testing
- Integrated contextual embeddings via a VectorDB layer, improving prompt-to-image accuracy by 50% overall
- Designed and deployed an auto-scaling GPU orchestration system with Kubernetes, halving average compute turnaround time

Undergraduate Researcher

September 2025 – Present

Allen Institute for AI

Seattle, WA

- Benchmarking reward models that align VLM outputs with human interaction for human in-the-loop training
- Collaborated with researchers to assess generalization of language context conditioned reward models across platforms
- Trained vision-language policies integrating reinforcement learning from human preferences (RLHF) to improve spatial reasoning and manipulation accuracy

Machine Learning Research Assistant

February 2025 – Present

University of Washington

Seattle, WA

- Developed prognostic multimodal deep reinforcement learning models to forecast cancer progression and survival
- Improved relative AUC accuracy by 23% over single-modal baselines, facilitating more effective disease treatment

AI/ML Applied Scientist Intern

June 2024 – September 2024

SabLab

Seattle, WA

- Built Python/PyTorch pipeline to clean and classify retinal imaging, causing 231% increase in labeled dataset size
- Developed transformer-based models to detect abnormal photoreceptor responses, 17% improvement over baselines

Technical Consultant

August 2025 – Present

Reading Partners

Remote

- Support technical operations for a nonprofit literacy program, integrating data tracking and automating workflows
- Worked alongside educators ensuring accurate, privacy-compliant student data management across regional centers
- Developed scripts and low-code tools to streamline tutor onboarding and reading-progress monitoring, cutting administrative time by 25%

PROJECTS

CRISPR Apples – 1st Place DubHacks 2025 | *Python, Gemini, OpenCV*

October 2024 – Present

- Largest hackathon in the north west with 1000+ participants, won \$2,000 cash + internship offer
- Built an AI-driven CRISPR gene-editing platform that generated and ranked guide RNAs using transformer-based scoring models, with Next.js + Tailwind interface over a Python inference backend

DubCalc | *Python, Manim, LaTeX*

October 2024 – Present

- Created dynamic visualizations to model limits, derivatives, and optimization concepts students struggle to grasp
- Accessed by ~50 students in our university's calculus class quarterly, improving engagement and learning

LEADERSHIP

Lead Math Tutor

September 2024 – Present

Math Study Center

University of Washington

- Built interactive visualizations and Python-based animations to demonstrate concepts; system developed into *DubCalc*
- Oversaw tutor on-boarding and training, leading groups in designing streamlined and consistent session workflows

AWARDS

Honors: Mary Gates Research Endowment, 1st Place DubHacks 2025, Study of Exceptional Talent, Dean's List (x7)